

AMIT MANJARLY

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TECHNICAL SKILLS

ML & Deep Learning: CNNs, ResNet, U-Net, Open-Set Recognition, Causal ML, Conformal Prediction, FNO / PINNs

LLMs & GenAI: Llama-3.2, Mistral, RAG, Fine-Tuning (QLoRA, rsLoRA), Hugging Face, ChromaDB

Computer Vision: YOLOv8/v11, OOD Detection, TorchGeo, Object Detection, Image Segmentation, CV Models

ML Frameworks: PyTorch, PyTorch Lightning, TensorFlow, Scikit-learn, XGBoost, LightGBM, NumPy, Pandas

Infra & MLOps: Docker, Kubernetes, MLflow, DVC, CUDA, Apple MPS, FastAPI, Streamlit, Google Earth Engine

Languages & DBs: Python (Advanced), SQL, C++; PostgreSQL, InsForge, SQLite, Hive, Impala, Oracle

PROFESSIONAL EXPERIENCE

Resilient

Jul 2026 - Present

Machine Learning Engineer Intern – Tempe, AZ

- Built Resilient Choco cocoa twin via **Sentinel-1/2 cross-attention rsLoRA adapters**, cutting weights 44.3x to 0.91 F1.
- Engineered **2D FNO physics** surrogates & climate-gated MoE ensemble for yields, achieving **0.85 mIoU** and 0.927 F1.
- Developed Resilient Arabica 730-day biennial coffee yield model & DynACof emulator, reaching low 0.1418 t/ha val RMSE.
- Deployed **TorchGeo segmentation** for coffee agroforestry mapping, achieving high-resolution **0.83 mIoU** & 0.9 F1 score.
- Architected causal **Deep R-Learners** & CQR conformal prediction intervals across eco-zones, attaining 90% PICP coverage.

Accenture PLC

Mar 2021 - Jul 2024

Software Engineer - Data & Backend – Hyderabad, India

- Built Python automated testing frameworks for global platforms, cutting manual QA pipeline cycle times by **40%**.
- Optimized mission-critical SQL data integration pipelines handling millions of records, reducing delays by 33%.
- Refactored core ETL scripts across Oracle and Hive databases to increase processing speed and eliminate friction sets.
- Partnered with analytics engineers to build Tableau dashboards for pipeline health tracking and prod dashboards setups.
- Solved workflow bottlenecks by analyzing system performance data, driving an aggregate **25% efficiency execution loops**.
- Automated data validation & quality checks in Python, improving downstream data reliability for data pipelines.
- Led Knowledge Transfer (KT) sessions for new engineers, speeding onboarding and standardizing ETL workflows.

PROJECTS

Open-World Continual Learning for Autonomous Agents [\[github\]](#)

Jan 2026 - Apr 2026

Product Optimization · Python · Architecture from Scratch · End-to-End Deployment

- Formulated EWC diagonal **Fisher Matrix** penalty ($\lambda=0.4$) to constrain parameter drift on θ^* during domain shift.
- Derived normalized **Shannon entropy scoring** $H(p)/\log(C)$ ($\text{thresh}=0.6$) on class logits to flag OOD obstacles as $\text{cls} = -1$.
- Mitigated rare-class gradient starvation by engineering 1:1:1:1 equalized prior class sampling across bounding labels.
- Evolved YOLOv8 to YOLOv11 vision backbones, stabilizing Distribution **Focal & CloU box regression loss** during transfer.

LLM Fine-Tuning for Mathematical Reasoning (STaR) [\[github\]](#)

Sep 2025 - Oct 2025

Large Language Models · Deep Learning · Strategic Roadmap · Iterative Improvement

- Fine-tuned language models utilizing bootstrapping to advance predictive text correctness and automated workflow runs.
- Analyzed baseline overfitting to define technical roadmaps, pivoting to cleaner data rationale loop architecture sets.
- Implemented 4-bit quantization to slash GPU footprints by 75%, delivering scalable, low-maintenance model system sets.

End-to-End Hybrid Recommendation Engine [\[github\]](#)

May 2025 - Jun 2025

Recommendation Systems · Predictive Modeling · Advanced Statistics · Metrics

- Built SVD collaborative filtering on MovieLens 100k, achieving Precision@10 of **0.26** vs. 0.15 baseline (**+73%**).
- Engineered a hybrid recommendation platform utilizing advanced statistics to drive 73% gain in user precision metrics.
- Built measurement pipelines via factor validation curves to track product performance against baseline metric profile.
- Blended collaborative and content filtering to solve cold-start constraints, maximizing user interaction density sets.

EDUCATION

M.S. in Data Science, Analytics and Engineering

Aug 2024 - May 2026

Arizona State University – Tempe, AZ

Coursework: Machine Learning, Deep Learning, NLP, Statistical Inference

B.E. in Computer Engineering

Aug 2015 - Jun 2019

Gujarat Technological University – Ahmedabad, India

Coursework: DSA, OS, DBMS, Computer Networks, OOP